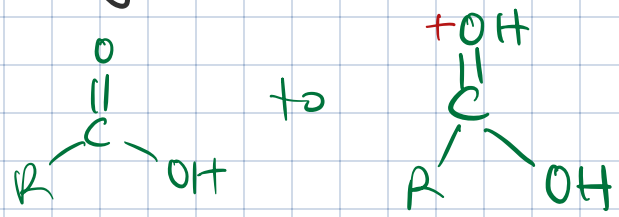
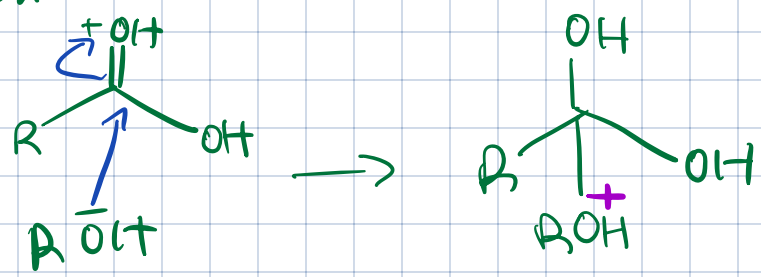
Fischer Esterification

- Acid catalyzed formation of Esters from a carboxylic acid and an alcohol.

1. Protonate



2. Carbonyl even more electron defficient, nucleophile R–OH attacks



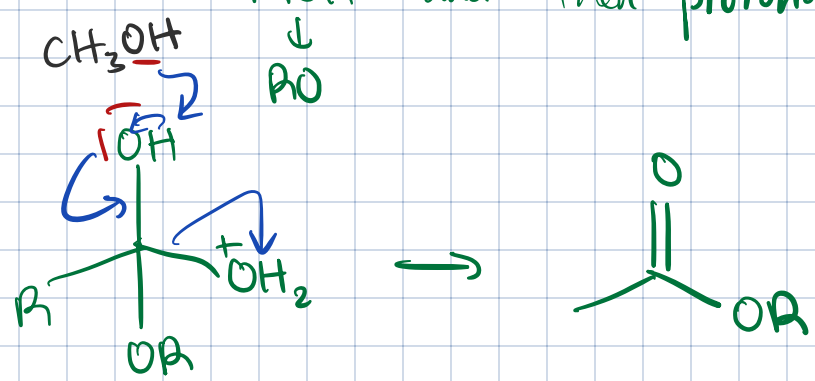
3. To reform the C=O, we must expel an OH and keep CH_3OH

↳ Use the solvent to deprotonate ROH^+ and then protonate OH^-

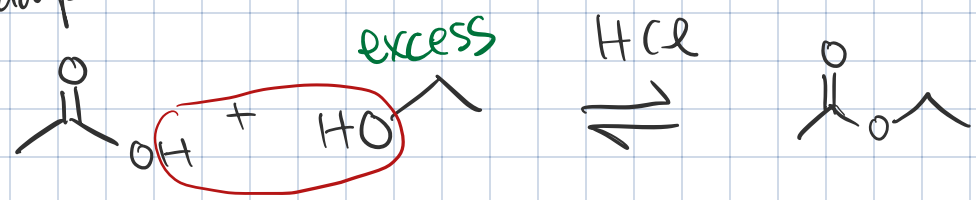
$ROH^+ \rightarrow RO$

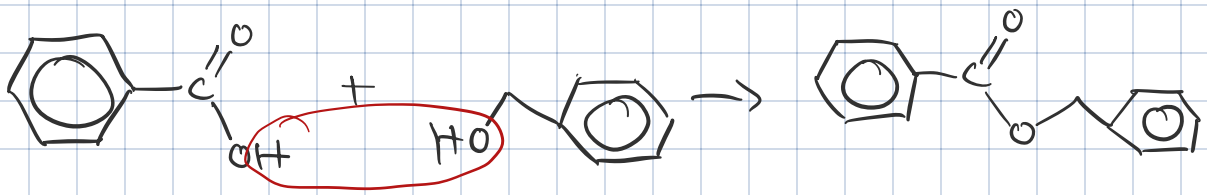
$OH^- \rightarrow H_2O$

4.

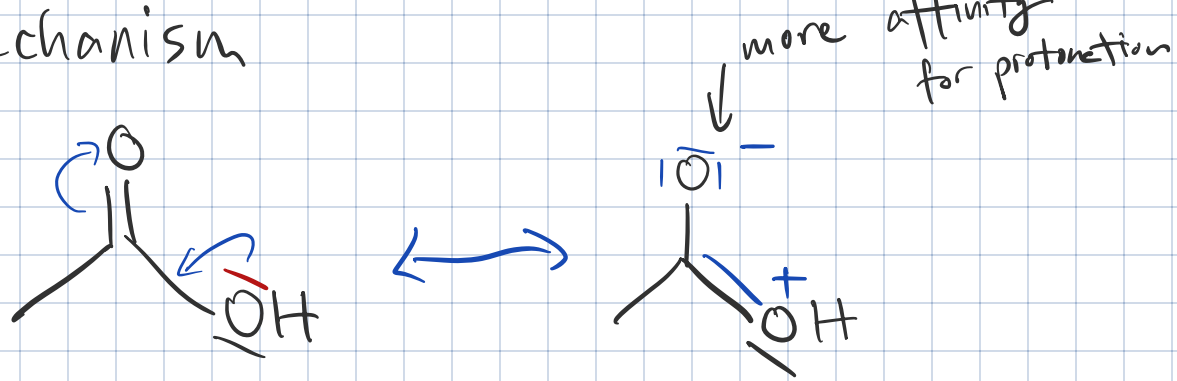


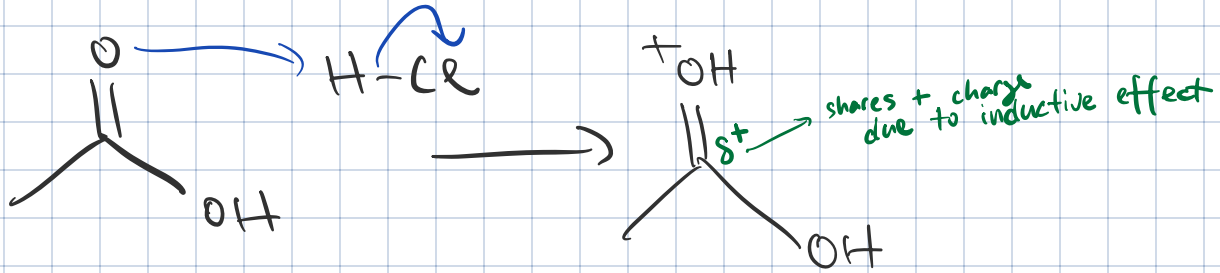
Examples

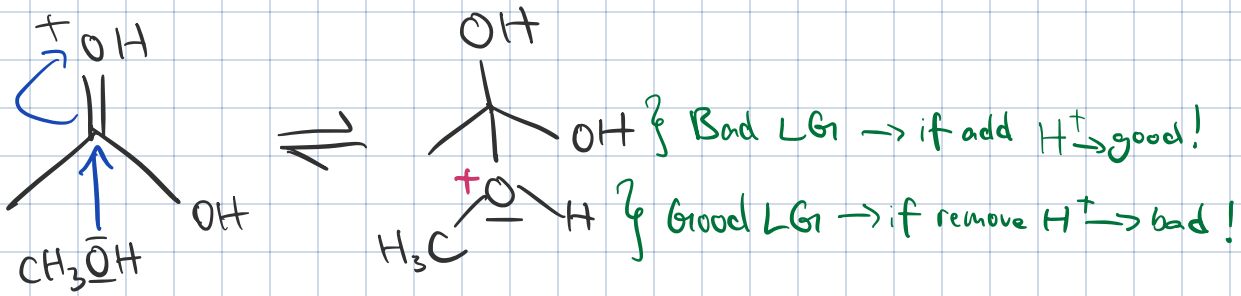


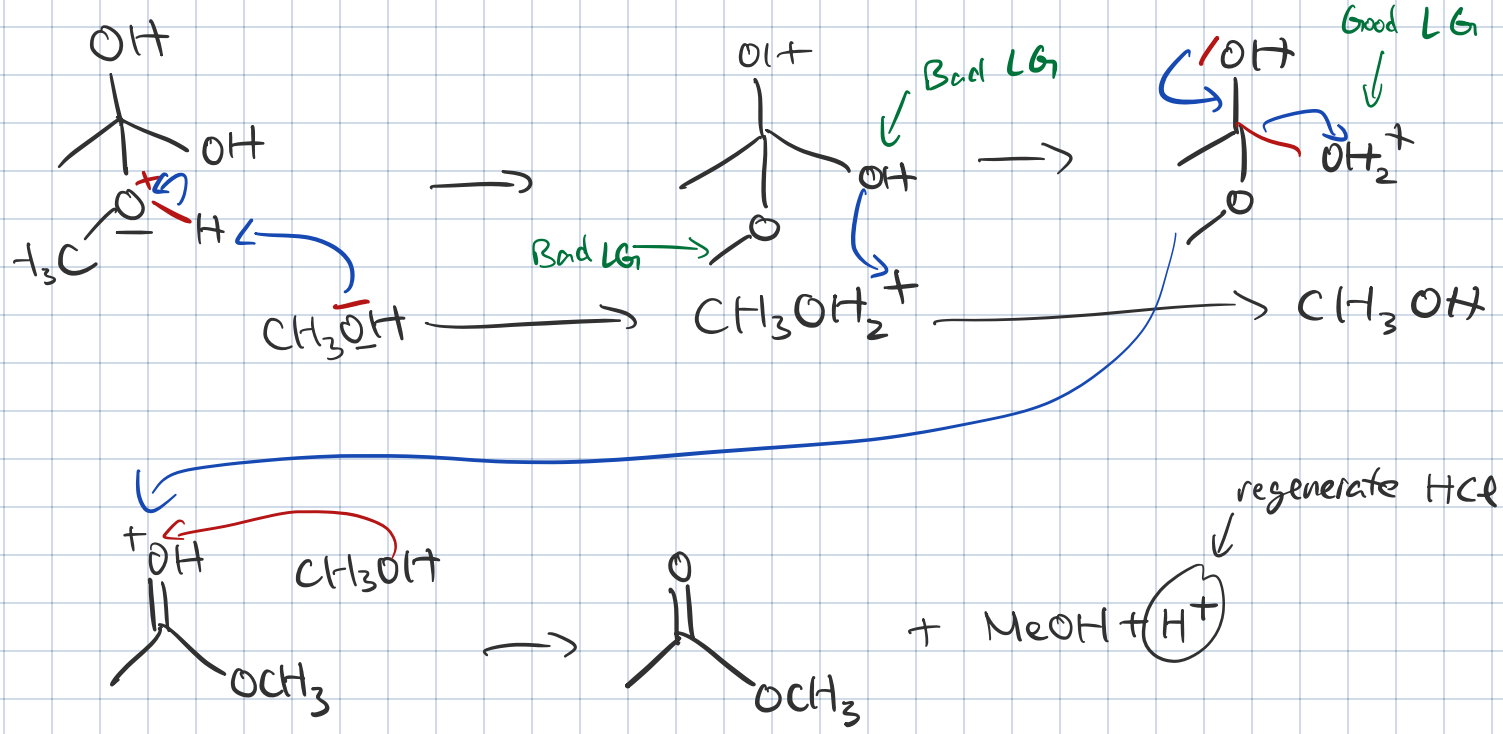


Mechanism





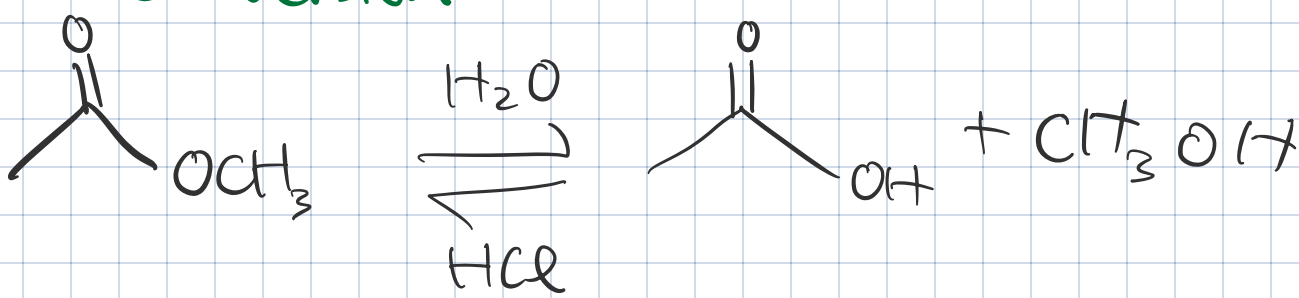




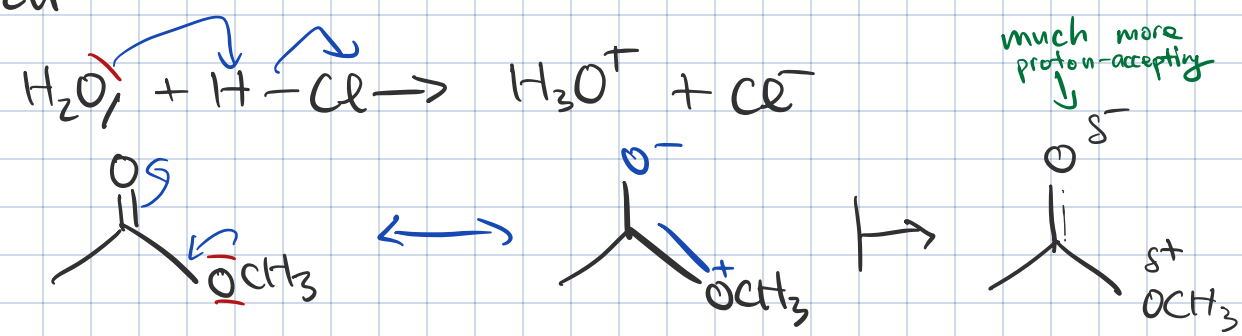
Ester Hydrolysis

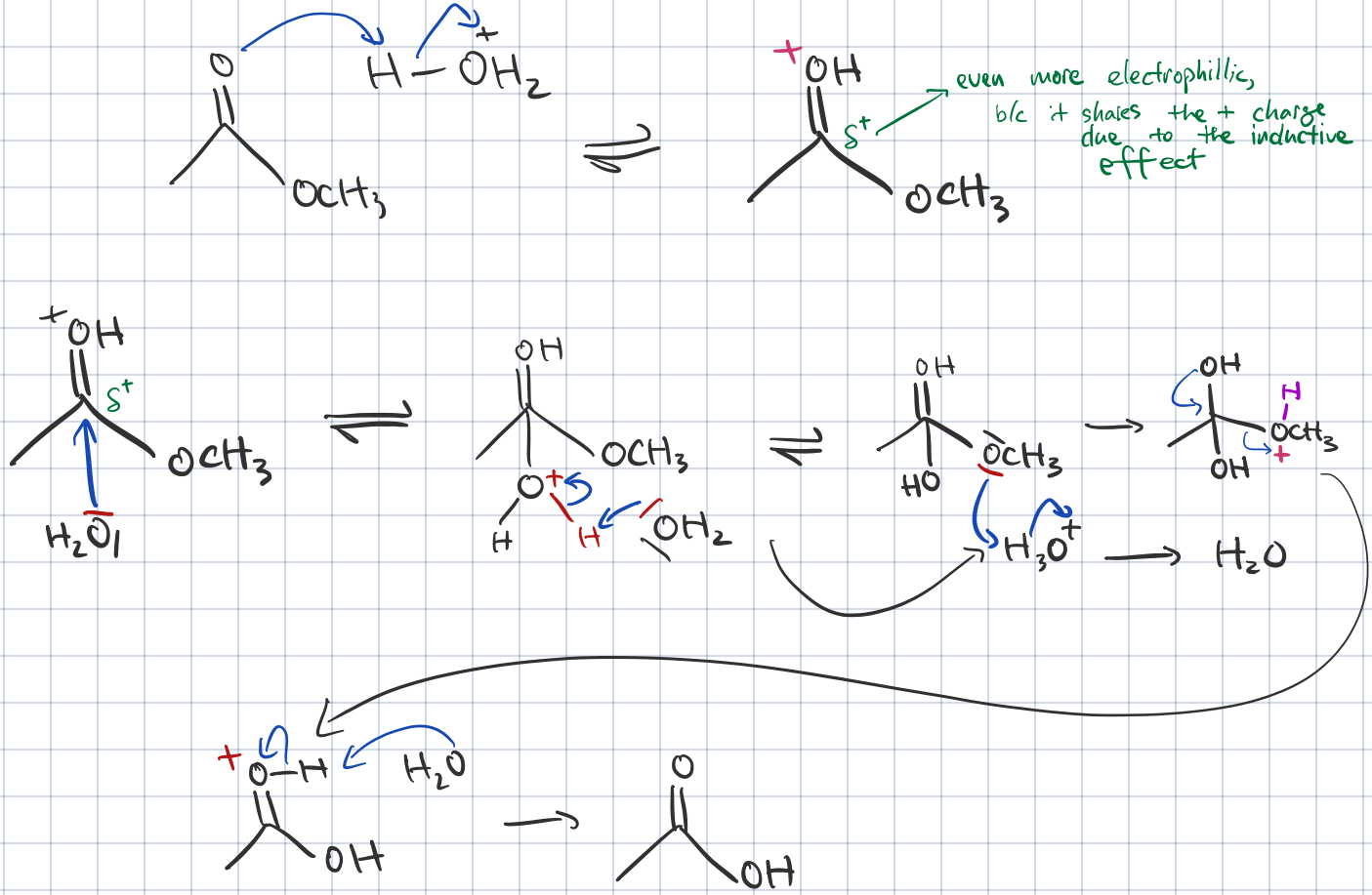
↳ Acid or Base catalyzed

Acidic Version

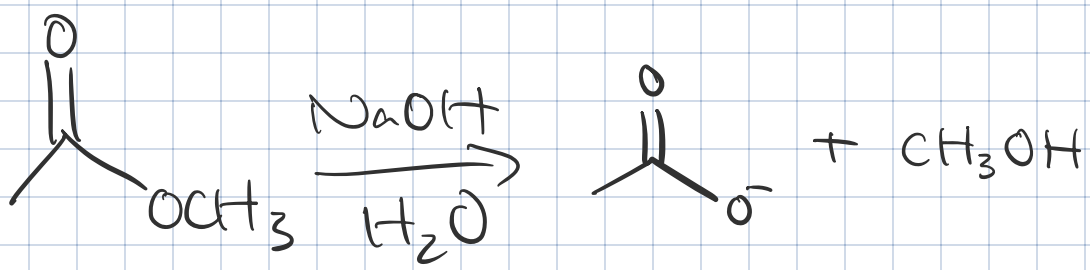


Mech:

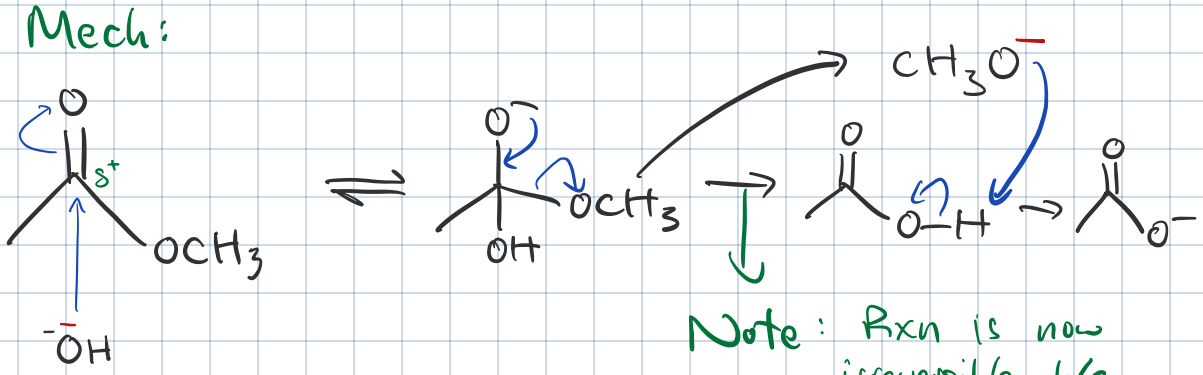




Basic Version



Mech:



Note: Rxn is now irreversible, b/c the CH_3O^- would much rather act as a base w/ the acidic carboxyl H than a Nuph w/ the C=O

Note: OH^- is not a catalyst. Although it speeds up the rxn, it is consumed, unlike the H^+ in the acid-catalyzed version